Classification of Gait Phases Using a Shank-Mounted Single IMU Sensor

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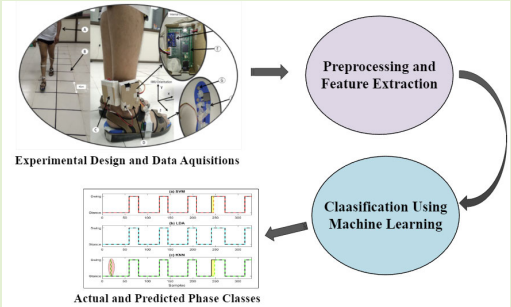
***Abstract*—Human gait parameters reflect physical and psychological health. Gait impairments hinder daily activities and mobility, particularly in those with neurological or musculoskeletal disorders. While methods for estimating temporal parameters exist, there remains a need for a portable, user-friendly, and cost-effective gait analysis device. Therefore, in this study, a feature engineering-based portable gait characterization module is proposed, and a shank-mounted inertial measurement unit (IMU) is utilized for gait phases (GPs) and event detection. The efficiency of the developed module was assessed using data collected from ten healthy subjects walking on a flat surface. A force-sensing resistor (FSR)-equipped instrumented insole was used as a reference system to validate the results obtained from the developed module. The performance is estimated using three different classifiers: support vector machine (SVM), K-nearest neighbor (KNN), and linear discriminant analysis (LDA). For gait event (GE) identifications, the average classification accuracies depicted by SVM, LDA, and KNN classifiers are 95.69% ± 5.23%, 96.64% ± 5.02%, and 93.63% ± 4.84% (p-value < 0.05), respectively. In summary, the GEs and gait temporal parameters are reliably estimated using a single IMU with SVM or LDA classifier (p-value > 0.05). Additionally, the efficacy of the proposed model for sensor location and subject variability has been evaluated. The performance of LDA and SVM classifier for GP prediction has been found to be invariant (p-value > 0.05) toward sensor location and subject variability. Future work will explore diverse walking conditions and test the module on individuals with gait impairments to evaluate its robustness and clinical applicability.**



***Index Terms*—Classification, embedded system, gait characterization, gait-phase (GP) prediction, inertial measurement unit (IMU).**

I. Introduction

GAIT analysis is a critical tool for assessing walking patterns and detecting irregularities indicative of abnormal physical or neurological conditions. Its applications span various fields, including clinical health diagnostics, rehabilitation, and sports performance enhancement [1]. In clinical diagnostics, gait analysis has been extensively used to monitor and evaluate ambulation in patients with Parkinson's disease (PD). Feature engineering techniques and advanced methodologies have been employed to distinguish gait patterns of individuals with PD from healthy controls [2], [3], [4], [5], [6], [7], [8], offering significant benefits for improving patient outcomes and quality of life.

In rehabilitation, gait analysis is instrumental in tracking recovery progress in patients with movement disorders, facilitating tailored interventions. In sports, it is utilized to optimize athletic performance by identifying biomechanical inefficiencies and preventing overuse injuries [9]. By addressing issues such as improper running form or faulty movement mechanics, athletes can enhance their performance and reduce the risk of injuries, including falls during critical phases like the mid-swing phase (SWP) of running. The growing need for real-time and portable gait monitoring systems has driven advancements in wearable and compact gait measurement devices, enabling seamless integration into real-life

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applications [10], [11]. These innovations hold promise for broadening the scope of gait analysis across healthcare and sports domains.

Gait characterization can be done using various techniques, from simple and affordable to highly advanced. For instance, footprint recording while ground walking to identify one's stride is cheap and straightforward. The shoe bottom is painted to accomplish the following so that the impression of the footprint can be observed on the ground. Solanki and Lahiri [11] and O'Sullivan et al. [12] employed standard walk tests, including the 6-min and 10-m walk tests, for gait analysis. The temporal metrics, such as step and stride times, are also measured during the ground walk using a stopwatch. Even so, this method of assessing gait is commonly used in standard settings though it still faces challenges such as limited resources and measurement subjectivity [11], [13]. The limitations and challenges in the conventional environment have been resolved with stereophotogrammetric systems [14], walk mats [15], [16], and wearable devices [17], [18], [19], [20]. The stereophotogrammetric system is a multicamera motion capture system that provides the instantaneous position of markers that can be used for gait characterization. While these systems excel in gait characterization, they are limited by their high costs, lengthy setup process, restricted to laboratory-based settings, and operational complexity, which arises from the specialized technical expertise required for operation. Similarly, pressure mats are more dependable and cost-effective than a stereophotogrammetric system for measuring gait, but the use is restricted to adequately equipped laboratories and overground gait exercises. These drawbacks motivated researchers to explore wearable and cost-effective solutions for gait characterization.

A. Related Works

Gait characterization can be performed using motion, force, goniometers, and piezoelectric sensors. Thus, researchers have started investigating the pressure sensors. The force-sensing resistor (FSR) has been utilized for gait characterization, as shown in the study conducted by Beauchet et al. [21]. FSRs placed beneath the feet are utilized to detect gait-related occurrences like toe-off (TO) and heel-strike (HS), enabling the measurement of spatiotemporal gait parameters. In the literature, researchers presented the significance and use of multiple FSRs (2–32 in number) located in different regions of the feet to characterize the gait [22], [23], [24]. Multiple FSRs improve the accuracy and precision of the measurement because they provides information on all parts of the feet touching the ground, which helps in detecting flat foot (FF) and gait abnormality. Nevertheless, it increases the number of data channels that improve the hardware complexity and difficulty identifying and rectifying faulty FSRs. This troubleshooting complexity renders it impractical for real-world application. Furthermore, few FSRs can fail to pick up specific components of gait impairment, such as foot eversion/inversion [25], often observed in poststroke subjects [26].

In prior research, researchers focused on developing practical and valid alternatives to address the constraints associated with laboratory testing, notably the time-consuming setup and costly equipment [27], [28]. The inertial measurement units (IMUs) appeared as a promising device to authorize unobstructed, affordable, daily life detection of gait in a wide abundance of cohabitation [29], ranging from neurological diseases such as somnambulism (sleepwalking) and Parkinson's [30]. Tomc and Matjačić [31] emphasized the benefits of adaptive frequency oscillators (AFOs) compared to conventional gait event (GE) detection techniques. They developed a real-time AFO-based approach to determine gait phase (GP) with a single head-mounted IMU and tested the method on a cohort of healthy individuals. Romijnders et al. [32] showed that shank-mounted IMUs can detect GEs during straight-line and curved walking in different neurological groups, using simple filtering methods. Luo et al. [33] introduced a new gait analysis algorithm for heel-mounted IMUs, employing refined techniques to precisely evaluate heel dynamics and compute a wide range of spatiotemporal parameters, along with measures of variability and symmetry. Experiments showed strong correlation ($r > 0.9$) and good agreement with gait parameters, with high reliability (ICC ≥ 0.921). The IMU sensor poses several challenges due to drift problems and inherent sensor noise. These challenges limited the widespread use of IMU-based systems for pervasive healthcare [34], [35], [36]. However, many literature presents reliable and efficient ways to estimate gait parameters using feature engineering techniques.

Machine learning (ML) enables automated gait analysis by detecting and classifying abnormalities from gait parameters. Hussain et al. [37] proposed an activity recognition system using a single gyroscope on the shank. Principal component analysis (PCA) eliminated redundant features, and classifiers such as K-nearest neighbor (KNN), support vector machine (SVM), and probabilistic neural networks were used. The model achieved 98.7% accuracy, outperforming existing systems. Camargo et al. [38] developed a locomotion mode classifier and environmental parameter estimator using ML and wearable sensors. The system achieved 99% accuracy in classifying ambulation modes and 96% in mode transitions, and accurately estimated ramp incline, stair height, and walking speed. Bailey et al. [39] developed an application using a consumer-grade smartwatch IMU. Forty-one young adults completed treadmill gait trials at three speeds, with single-stride outcomes and variability recorded via an optoelectronic system and 232 IMU metrics captured with an Apple Watch. These metrics trained models (linear, ridge, SVM, random forest, and xGB) for spatiotemporal outcomes. Tiwari et al. [3] identified level ground and ramp terrains for damping in prostheses using foot-to-ground angle (FGA). Data from Healthy ($n = 5$) and amputee ($n = 1$) participants showed 79.57% accuracy (Healthy) and 73.06% (amputee) with SVM, and 83.45% and 80% with CNN. Dehzangi et al. [40] proposed a gait identification method using time–frequency expansion and DCNN for feature extraction. Sensor data from five locations showed 91% accuracy with individual IMUs, improving to 97.06% with late sensor fusion. The study conducted by Hannink et al. [41] utilized deep learning to calculate the stride length and trained the model using a freely available dataset of 101 geriatric patients. The research did not prioritize the assessment of temporal gait metrics. Moreover, deep neural networks are very complex for implementation

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and online gait estimation for real-time embedded applications [3], [40], [42].

B. Contributions

In this study, a single IMU sensor-based system has been developed, and its performance has been validated against an FSR sensor insole system [22], [23], [43]. In addition, temporal GEs and phase estimation approaches have been proposed using feature engineering techniques. This fully data-driven approach does not depend on assumptions or theoretical models. Furthermore, the estimation of GEs (TO and HS) was performed using GP information [SWP and stance phase (STP)], mainly calculated using an IMU sensor signal. The main contributions of this article are listed as follows.

1) *Single IMU for Gait Parameter Estimation:* Unlike conventional methods requiring multiple sensors or complex setups, our approach utilizes a single shank-mounted IMU to accurately estimate GEs and temporal parameters, making it simpler and more practical for real-world applications.
2) *Portable and Cost-Effective Design:* By eliminating the need for expensive motion capture systems or instrumented setups, our system offers a user-friendly, cost-efficient alternative suitable for both clinical and nonclinical environments.
3) *Robust Gait Detection and Classification:* The data-driven method features advanced engineering for precise GE detection, surpassing standard IMU studies. It also achieves high accuracy with SVM and linear discriminant analysis (LDA) classifiers, ensuring adaptability across sensor placements and subjects.
4) The robustness of the proposed model is presented by estimating the performance from the different locations of the sensor.
5) *Comparison With Existing Studies:* The comparison of the proposed method with existing techniques has been presented.
6) *Real-Time Implementation Potential:* The computational efficiency of the approach enables seamless real-time integration, paving the way for practical applications in gait monitoring and rehabilitation.

II. Materials and Methods

A. Gait Temporal Characterization

Human gait is the linear advancement of the human body's movement generated by synchronized, rotated movements of body parts. The fundamental purpose of gait is to support the head, arms, and trunk (HAT), balance the body, and maintain an upright posture. Gait is defined as a balanced, alternating, and cadenced movement of lower limbs that results in the forward progression of the body. A stride represents the single gait cycle (GC), which begins with HSs and ends when the next HSs are on the ground of the same leg. The single GC contains a 60% STP and a 40% SWP [44]. The STP contains subevents such as HS, heel-lift (HL), mid-stance (MS), FF, and TO. Further, the SWP is divided into the initial, mid, and terminal swings, as shown in Fig. 1.

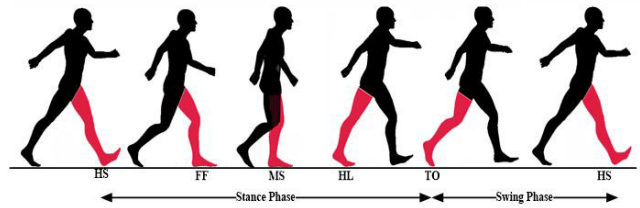

Fig. 1. GPs and GEs of a complete GC [45].

Gait temporal parameters include stride time, step time, stance time, single support time, cadence, and swing time [46]. This study focuses on four parameters: stride time, step time, stance time, and swing time. Furthermore, these parameters are used in identifying gait impairment, assessing, and treating individuals with pathological situations that affect their walking, such as Parkinson's [30] and Stroke [47]. Any movement disorder affects the walking of the patients, which could be identified using these parameters. The GC can be subdivided into two phases: one STP and a second SWP. HS of any leg represents the starting of the STP; however, the SWP initiates when TO happens for the same leg. If the TO and HS of the GC are detected, then all spatiotemporal parameters are measured. Equations (1)–(4) outline the methodology for calculating various temporal gait parameters, providing a detailed framework for their measurement and analysis [48].

1) Stride Time: It is the duration between two consecutive HSs of the same foot and is calculated using the time between HS(i) of one leg and the next HS($i+1$) of the same leg

$$\text{Sride}(i) = \text{HS}(i+1) - \text{HS}(i). \tag{1}$$

2) Stance Period: It is the time duration between HS and TO of the same leg and calculated using the time between TO(i) and HS(i) of the same leg

$$\text{Stance}(i) = \text{TO}(i) - \text{HS}(i). \tag{2}$$

3) Swing Period: This refers to the period during which the foot is completely lifted off the ground, which is calculated by measuring the time between TO(i) and HS($i+1$)

$$\text{Swing}(i) = \text{HS}(i+1) - \text{TO}(i). \tag{3}$$

4) Step Time: This is the duration between two consecutive foot contacts with the ground of the opposing foot. It is calculated using the time between left leg HS(i) and right leg HS(i)

$$\text{Step Time}(i) = \text{HS}_{\text{right}}(i) - \text{HS}_{\text{left}}(i). \tag{4}$$

B. Developed Hardware System

The proposed module consists of four submodules: one 6-degrees of freedom (DoFs) IMU sensor, a power supply, an instrumented insole with FSR sensors, and the microcontroller, as shown in Fig. 2. The microcontroller has 4k electrically erasable programmable read-only memory (EEPROM) (emulated) memory, 1024k RAM, 7936k Flash, ARM Cortex-M7 at 600 MHz, eight serial, three serial peripheral interface (SPI), three interintegrated circuit (I2C) ports, 18 analog input pins, 35 pulsewidth modulation (PWM) output pins, 55 digital input/output pins,

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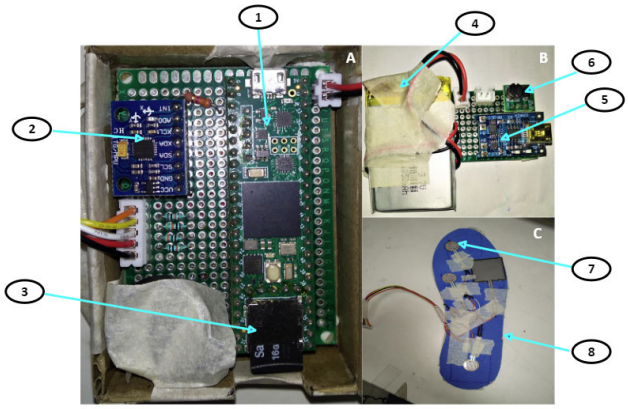

Fig. 2. Developed hardware module. (1) Microcontroller. (2) IMU. (3) SD card. (4) Li-ion battery. (5) Battery charging module. (6) DC–dc voltage regulator. (7) FSR sensor. (8) Insole. (a) Main module, (b) battery Module, and (c) instrumented insole with FSR sensors.

TABLE I
DEMOGRAPHIC INFORMATION OF STUDY PARTICIPANTS

ID	Age (years)	Sex	Weight (Kg)	Height (cm)	BMI (kg/m2)
H01	27	M	70.00	183.00	20.90
H02	35	M	78.00	179.00	24.30
H03	30	M	80.00	176.00	25.59
H04	31	M	67.00	160.00	26.17
H05	26	M	72.00	160.00	27.95
H06	27	M	86.00	187.00	24.59
H07	21	M	85.00	174.00	28.07
H08	26	M	60.00	165.00	22.03
H09	25	M	69.00	170.00	23.73
H10	21	M	73.00	175.00	24.03

SD card port, and real-time clock (RTC) for date/time. The Arduino IDE software, supplemented with the Teensyduino add-on, serves as the programming environment for this microcontroller. The insole consisted of four FSR sensors [21], one at the toe, one at the heel, and two at the metatarsal region of the foot. The IMU sensor is linked to the microcontroller's I2C digital pins. This IMU unit measures a three-axis gyroscope (Gyx, Gyy, and Gyz) with a three-axis accelerometer (Accx, Accy, Accz) to form a standalone six-axis unit. The IMU sensor is mounted in the main module, as depicted in Fig. 2. To perform the data acquisition and detect gait walking patterns, a module is affixed to the shank area of the lower limb, positioned above the ankle. The developed module consumes 125 mW of power when operating at full speed (600 MHz) with a 100 mA current and 1.25 V supply. The battery used has a capacity of 1000 mAh, so the total duration it can run for (1000 mAh/100 mA) is 10 h. The output signals from FSR and IMU had a sampling frequency of 100 Hz. Simultaneously, these signals were stored on the data storage module (SD card) in the text file.

C. Experimental Setup and Data Acquisition

The developed hardware was used to acquire the signal for walking activity, which was further utilized for ML model development and gait analysis. Table I presents the demographic information of study participants. A total of ten participants, without any neuromuscular and gait disorder (average height of 173.08 ± 9.11 cm and a weight of 74.06 ± 8.21 kg) aged between 20 and 40 years, voluntarily participated in the experimental sessions. The Indian Institute of Technology (IIT) Delhi Ethics Committee granted approval for the proposed research (ref no: 2021/P015). The subjects were asked to give written consent before the experiment. The signals acquired during the experiment were purely noninvasive. The signals were acquired using the developed sensor module, which consists of an FSR sensor and an IMU sensor. The sensor module developed was affixed to the shank area of the participants' right foot, specifically at the ankle, as shown in Fig. 3 [43].

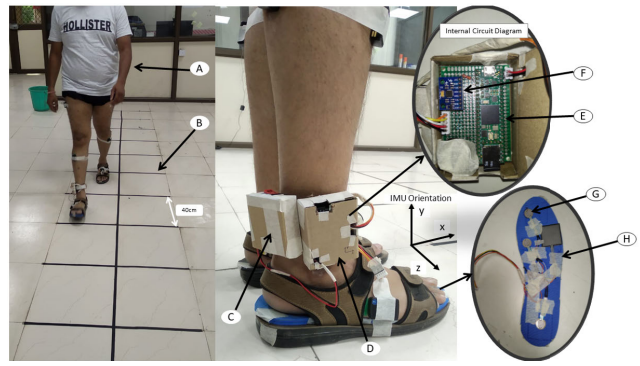

Fig. 3. Experimental setup and wearable device. (a) Participant. (b) Walking path. (c) Battery module. (d) Central module with IMU. (e) Teensy 4.1 microcontroller with SD card. (f) IMU sensor. (g) FSR sensor. (h) Insole.

The wearable device is sized 7.5 × 5.5 × 2.5 cm. The inertial sensor is oriented so that the x-axis aligns with the walking direction. The complete experimental setup is depicted in Fig. 3. The laboratory featured a straight path measuring 10 m in length and 0.8 m in width, marked with one-sided black tape.

In the experiment, the participants were asked to walk at a self-paced (on an average of 1.2 m/s) on the experimental pathway. For each subject, a total of ten trials were recorded; further, the subjects were instructed to flex his/her knee with a full range of motion, followed by foot tapping, at the start and end of each trial. It was used to synchronize the IMU and FSR signals and to separate the gait trials. The collected data during the experiment were stored in the SD card module of the main system.

D. Signal Processing

All the signal processing was implemented with MATLAB R2021b, on a computer with Windows 10, an Intel[1] Core i9-10900K CPU at 3.70 GHz and 128-GB RAM. The acquired FSR and IMU signals were stored in an SD card, and data were imported to the PC for further analysis. Fig. 4 shows the normalized raw signals of IMU and FSR sensors (heel and toe).

1) GP and GE Detection Using FSR: The signal of two FSR sensors (heel and toe) was utilized to estimate GPs (STP and SWP) and GEs (HS and TO) and to estimate the gait temporal parameters, as shown in (1)–(4) [22], [23]. Moreover, these values also serve as a reference to estimate the performance

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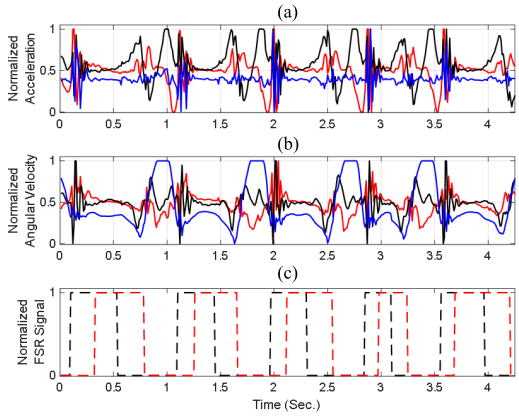

Fig. 4. Normalized raw signals of IMU and FSR sensor. (a) and (b) *X*-axis: black, *Y*-axis: red, and *Z*-axis: blue. (c) Heel FSR: black dotted and toe FSR: red dotted.

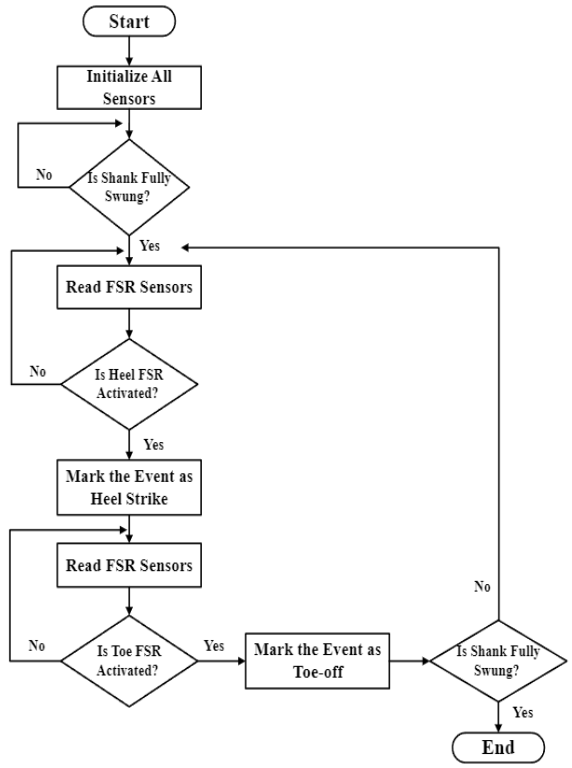

Fig. 5. Flowchart and pseudocode to detect the GEs and GPs from the FSR signal.

of the developed IMU-based gait characterization model. First of all, the walking trials were separated using the IMU sensor signal. Fig. 4(a) and (b) shows the recorded IMU signal of subject 1. As mentioned earlier, the subjects were asked to flex his/her instrumented leg's knee twice at the start and end of each trial. These events are distinctively identified by the z-axis gyroscope (Gyz) signal; hence, it was used to separate the individual trials. Further, for every trial, TO and HS events have been identified using the FSR dataset.

Fig. 5 and Algorithm 1 show the flowchart and algorithm to detect the GEs and GPs from the FSR signal. At every instance, the designed algorithm looked for the knee flex event (annotating the start of the trial) as soon as it was detected, and the algorithm checked for the HS and subsequently for the TO event or the next knee flex event (annotating the end of the trial). The TO and HS events detected in between the trials were used to estimate the GP and gait temporal parameters. These temporal gait parameters and GPs served as reference values for IMU-based model validation.

2) Feature Extraction and Classification Algorithm: Fig. 6 illustrates a block diagram representing the signal processing pipeline employed in the design of the gait characterization module utilizing the IMU signal. All the signals were imported from the SD card to the computer. During the preprocessing stage, the IMU signal from all 6-DoF was filtered using a zero-lag, fourth-order Butterworth bandpass filter, with a cutoff frequency range of 0.5–10 Hz [49]. Moreover, filtered data were normalized through the min–max normalization method, which ensures consistency and comparability across different datasets [50]. Additionally, the precise extraction trials from recorded data are a crucial step for the performance of the training model. The trials were extracted using the Gyz signal generated from voluntary knee flexion movement performed at the start and end of the trial. Subsequently, an overlapped windowing algorithm is utilized to segment all the signals (gyroscope and accelerometer) for further processing. For this study, a window size and window shift of 15 and one sample were opted for, respectively. Further, for each window, six-temporal features were extracted, as presented in Table II. The extracted features were skewness, variance, root mean square, kurtosis for individual DoF, and correlation between each pair of DoF of gyroscope and accelerometer [4]. It resulted in a total of 30 features for each feature vector. The feature vector associated with the SWP was designated as "1," while those belonging to the STP were labeled as "2." The system's performance was assessed using tenfold cross-validation. In this process, out of ten recorded trials, one trial was reserved for testing the classifier, while the remaining trials were utilized for training. This cross-validation procedure was repeated until each trial underwent testing.

Additionally, the computation of six performance indices, including precision, accuracy, sensitivity, Matthews correlation coefficient (MCC), Cohen's kappa, and $F1$-score, provides a multifaceted evaluation of classifier performance, allowing for nuanced comparisons and insights into their capabilities. The computation of multiple performance indices is crucial in evaluating the effectiveness and reliability of classifiers in motion analysis. Accuracy, as a fundamental measure, indicates the overall correctness of the classifier's predictions, providing a broad assessment of its performance. Precision emphasizes the ratio of true positive predictions to all positive predictions, offering insights into the classifier's ability to avoid false positives. The sensitivity of a prediction system is defined as the percentage of correct predictions relative to the total number of positive instances, highlighting the classifier's capability to detect positive cases accurately. The $F1$-score is a harmonic average of precision and sensitivity, balances these two metrics, and provides a comprehensive evaluation of the classifier's overall performance. Cohen's kappa statistic assesses the agreement between the classifier's predictions and

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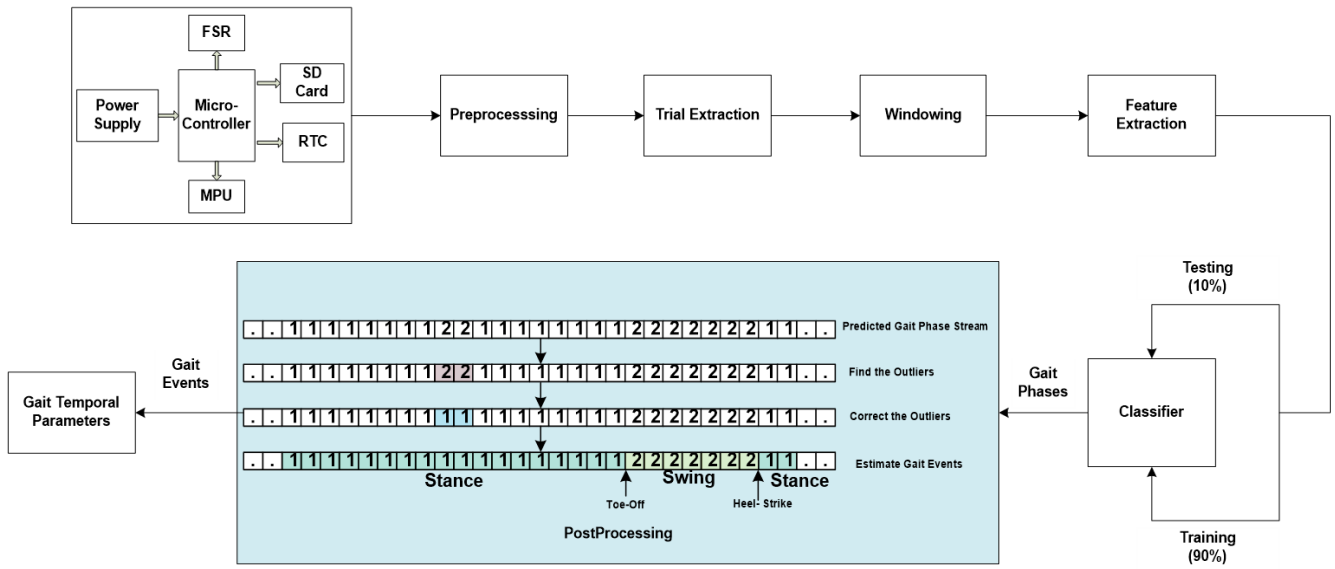

Fig. 6. Signal processing pipeline utilized for gait characterization module design using the IMU signal.

Algorithm 1 Extracting GEs, GPs, and Temporal Gait Parameters From the FSR Signal: Benchmark Data

Input: Angular velocity from shank worn IMU signal: Gyz, FSR sensor signal: FSRtoe, and FSRheel

Output: Gait event: HS and TO, Gait Phases: Swing and Stance, Temporal gait parameters: Stance period, Stride time, Swing period and Step time

```
Start                                          ▷ initiation of trial
if (Gyz ≥ Gyz_max) then
    start=1
    While (start ==1)                          ▷ within the trial
    if FSRheel == 1 then                       ▷ check for the heel strike
        HS(i) =current time stamp              ▷ record the heel strike time
        if FSRtoe == 1 then                    ▷ check for the toe-off
            TO(i) =current time stamp.         ▷ record the toe-off time
        elsewait for FSRtoe == 1               ▷ wait for toe-off
        end if
    elsewait for FSRheel== 1                   ▷ wait for heal strike
    end if
    if (Gyz ≥ Gyz_max) then                    ▷ end of trial
        start = 0.
        goto step 20
    elsegoto step 4
    end if
    end
elsewait (Gyz ≥ Gyz_max)                       ▷ wait for initiation of trial
end if
Estimate all the gait temporal parameters using eq (1)-(4)
Stop
```

TABLE II
FEATURES EXTRACTED FROM IMU SIGNALS

#	Feature	Acceleration Signals (AccX, AccY, AccZ)
F1-F3	Variance (X,Y,Z)	$VAR = \frac{1}{L-1}\sum_{i=1}^{L} Acc_i^2$
F4-F6	Skewness (X,Y,Z)	$SKW = \sum_{i=1}^{L} \frac{(Acc_i - Acc_{mean})^3}{(\sqrt{VAR})^3}$
F7-F9	Root mean square (X,Y,Z)	$RMS = \sqrt{\frac{1}{L}\sum_{i=1}^{L} Acc_i^2}$
F10-F12	Kurtosis (X,Y,Z)	$KRT = \sum_{i=1}^{L} \frac{(Acc_i - Acc_{mean})^4}{(\sqrt{VAR})^4}$
F13-F15	Linear Correlation (X-Y, X-Z, Y-Z)	Cor(x,y)= $\frac{\sum_{i=1}^{L}(X-X_{mean})(Y-Y_{mean})}{\sqrt{\sum_{i=1}^{L}(X-X_{mean})^2(Y-Y_{mean})^2}}$
#	Feature	Gyroscope Signals (GyX, GyY, GyZ)
F16-F18	Variance (X,Y,Z)	$VAR = \frac{1}{L-1}\sum_{i=1}^{L} Gy_i^2$
F18-F21	Skewness (X,Y,Z)	$SKW = \sum_{i=1}^{L} \frac{(Gy_i - Gy_{mean})^3}{(\sqrt{VAR})^3}$
F22-F24	Root mean square (X,Y,Z)	$RMS = \sqrt{\frac{1}{L}\sum_{i=1}^{L} Gy_i^2}$
F25-F27	Kurtosis (X,Y,Z)	$KRT = \sum_{i=1}^{L} \frac{(Gy_i - Gy_{mean})^4}{(\sqrt{VAR})^4}$
F28-F30	Linear Correlation (X-Y, X-Z, Y-Z)	Cor(x,y) = $\frac{\sum_{i=1}^{L}(X-X_{mean})(Y-Y_{mean})}{\sqrt{\sum_{i=1}^{L}(X-X_{mean})^2(Y-Y_{mean})^2}}$

the ground-truth labels, considering the possibility of random agreement, thus offering a robust measure of classification reliability. Whereas, the MCC provides a balanced assessment of classifier performance even in imbalanced datasets. These performance measures offer a nuanced understanding of classifier capabilities, allowing researchers to make informed decisions regarding their suitability for motion analysis tasks [51].

The output stream of the detected GP from each classification method was subjected to a "postprocessing algorithm" with the objective of identifying and eliminating outliers that were embedded in larger sections that belonged to the

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Algorithm 2 Estimating GEs, GPs, and Temporal Gait Parameters From IMU Signal

Input: Acceleration and Angular velocity from shank worn IMU signal: Accx, Accy, Accz, Gyx, Gyy and Gyz
Output: Gait Phases: Swing and Stance, Gait event: HS and TO, Temporal gait parameters: Stance period, Stride time, Swing period and Step time

```
Start                                                                   ▷ initiation of trial
for each trial do
    Preprocess all the signals of IMU                                   ▷ Filtering
    for each window (window size=15 sample and window shift=1 sample) do ▷ Extract the signal window of 15 sample
        Extract the feature for IMU signal                              ▷ Total six features for each signal have been extracted
        Construct the feature vector using all the features             ▷ Feature will have total 30 features
        Label the feature vector (as per the STP/SWP) and add to dataset ▷ Labeling will be done as per FSR signal
    end for
end for
Divide the dataset in subsets as per 10-fold cross-validation           ▷ Prepare the dataset for cross-validation
for each subset of dataset do                                           ▷ Estimate the performance of classifiers for each subset
    Train and test the classifiers for predicting the gait phase (stance/swing) ▷ Three classifiers have been compared
    Postprocess the predicted gait phase stream to filter the outliers
    Evaluate the performance indices                                    ▷ performance indices for classifier performances
    Estimate the gait events (HS/TO) from the postprocessed gait phase stream
    Estimate all the gait temporal parameters using eq (1)-(4)
end for
Stop
```

TABLE III
LEARNING PARAMETERS FOR VARIOUS CLASSIFIERS

Classifiers	Learning Parameters
SVM	Kernel Type = RBF; Coding-one vs one; optimum-$\delta^2 = 1.9, optimum - C = 7, Poly - order = 2, 3, 4$
LDA	Discrimiype: "Linear"; Delta: 0.02; Gamma: 0.0003
KNN	NumNeighbours: 3; Distance: 'Euclidean'

other class. When a cluster was less than half a window (i.e., seven samples or 0.07 s) and preceded and followed by a larger, differently categorized array, the cluster was said to be an "outlier" (Fig. 6). The current class is "preferred" by the correction process, which operates consecutively. Short clusters designated as "swing" are changed into "stance" if we are in a "stance" phase. Further, from the corrected GP stream, GEs were found by detecting the transitions from stance to swing (TO event) and swing to stance (HS event). This postprocessing algorithm improves the proposed model performance in detecting the GPs and GEs and so as the gait temporal parameters. For this study, the performance of three different classifiers, KNN, LDA, and SVM, were evaluated and compared for the individual subject. In addition, an ML toolbox in MATLAB (R2020b) was utilized to find the optimum parameters of SVM, LDA, and KNN. Table III presents the learning range and the optimum values of parameters for various classifiers. Moreover, the complete algorithm for estimating temporal gait parameters using a single IMU signal is comprehensively presented in Algorithm 2. This includes the detailed methodology for capturing, processing, and analyzing the IMU data to accurately extract different phases and events of gait, ensuring reliable estimation of temporal parameters.

III. RESULTS

In this work, the ML model has been implemented to first evaluate the GPs (swing and stance), and then using the phase information, the GEs (TO and HS) and temporal gait parameters have been estimated using a single IMU signal. The performance of three different classifiers has been compared comprehensively for ten subjects. For each subject, a total of ten walking trials have been recorded, and from each walking trial, five GCs have been used for dataset preparation. The GP detection problem has been considered a two-class classification problem.

Table IV displays the diverse evaluation metrics of the GP detection model across individual subjects and all classifiers. The % classification accuracy of the SVM classifier ranges between 80.37% and 99.29%, for the LDA classifier, it is between 89.63% and 99.26%, and for the KNN classifier, it is found between 72.20% and 98.73%. Whereas the average % classification accuracy has been found as 95.68% ± 5.22%, 96.64% ± 5.02%, and 93.62% ± 4.83% (p-value < 0.05) for SVM, LDA, and KNN classifiers, respectively. The proposed model performance is equally dependent on the correct classification of stance as well as the SWP. Hence, to compare the performance of classifiers more comprehensively, precision, sensitivity, $F1$-score, C_{kappa}, and MCC are also estimated. It will provide more insight into the results obtained using the developed classification model. The average % precision has been found as 96.58 ± 4.26, 97.72 ± 3.81, and 96.47 ± 3.80 (p-value > 0.05) for SVM, LDA, and KNN classifiers, respectively. The average % recall has been reported as 97.32 ± 3.85, 97.45 ± 3.99, and 94.31 ± 4.25 (p-value < 0.05) for SVM, LDA, and KNN classifiers, respectively. Further, by analyzing the performance of the classifier in view of sensitivity and recall more closely, $F1$-score has been estimated. The average % $F1$-score has been found as 96.91 ± 3.78, 97.54 ± 3.71, and 95.32 ± 3.61 (p-value < 0.05) for SVM, LDA, and KNN classifiers, respectively.

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TABLE IV
PERFORMANCE OF DIFFERENT CLASSIFIERS FOR GP DETECTION

SVM						
Subject No.	**Accuracy**	**Precision**	**Sensitivity**	**F1 score**	C_{kappa}	**MCC**
S1	91.37 ± 18.14	93.27 ± 12.30	94.36 ± 14.46	93.71 ± 13.21	87.34 ± 19.96	89.07 ± 15.75
S2	97.61 ± 3.44	97.12 ± 5.05	99.57 ± 0.88	98.26 ± 2.64	94.28 ± 7.59	94.66 ± 6.92
S3	98.82 ± 2.11	99.26 ± 1.03	99.23 ± 1.82	99.24 ± 1.29	96.44 ± 7.15	96.49 ± 7.06
S4	98.11 ± 2.55	98.06 ± 3.10	99.23 ± 1.21	98.62 ± 1.87	95.66 ± 5.85	95.78 ± 5.59
S5	97.93 ± 2.69	98.66 ± 0.88	98.53 ± 3.57	98.56 ± 1.88	94.89 ± 6.54	95.09 ± 6.06
S6	97.92 ± 1.88	97.81 ± 2.84	99.23 ± 1.15	98.49 ± 1.39	95.17 ± 4.28	95.33 ± 4.04
S7	98.19 ± 1.67	99.08 ± 1.22	98.31 ± 2.02	98.68 ± 1.23	95.79 ± 3.83	95.86 ± 3.77
S8	80.37 ± 14.88	85.68 ± 11.12	86.63 ± 11.18	86.08 ± 10.80	55.65 ± 29.02	56.34 ± 28.86
S9	97.29 ± 3.51	97.68 ± 3.82	98.47 ± 1.44	98.04 ± 2.49	93.64 ± 8.39	93.76 ± 8.15
S10	99.29 ± 1.39	99.24 ± 1.31	99.71 ± 0.88	99.47 ± 1.07	98.40 ± 3.04	98.41 ± 3.04
LDA						
Subject No.	**Accuracy**	**Precision**	**Sensitivity**	**F1 score**	C_{kappa}	**MCC**
S1	91.24 ± 20.60	93.74 ± 14.01	93.13 ± 17.61	93.30 ± 15.90	89.12 ± 19.36	91.29 ± 13.58
S2	96.68 ± 4.03	96.24 ± 5.41	98.83 ± 1.86	97.44 ± 3.20	92.74 ± 8.58	93.04 ± 8.13
S3	99.26 ± 1.99	99.67 ± 0.69	99.42 ± 1.75	99.54 ± 1.21	97.58 ± 6.67	97.62 ± 6.55
S4	98.42 ± 2.15	98.53 ± 2.60	99.23 ± 1.37	98.86 ± 1.52	96.29 ± 5.18	96.42 ± 4.91
S5	99.12 ± 2.05	99.75 ± 47	99.09 ± 2.74	99.40 ± 1.41	97.77 ± 5.12	97.90 ± 4.75
S6	98.91 ± 1.18	99.42 ± 0.76	99.02 ± 1.40	99.22 ± 0.82	97.42 ± 2.97	97.46 ± 2.91
S7	98.76 ± 1.76	99.54 ± 0.87	98.66 ± 2.06	99.09 ± 1.29	97.15 ± 4.06	97.20 ± 3.99
S8	89.63 ± 9.65	94.09 ± 7.35	91.37 ± 7.21	92.66 ± 6.94	74.81 ± 22.96	75.13 ± 23.04
S9	96.01 ± 4.57	96.86 ± 5.09	97.53 ± 1.48	97.15 ± 3.24	90.52 ± 10.89	90.72 ± 10.57
S10	98.36 ± 2.38	99.37 ± 0.90	98.29 ± 2.46	98.82 ± 1.66	96.13 ± 5.34	96.18 ± 5.25
KNN						
Subject No.	**Accuracy**	**Precision**	**Sensitivity**	**F1 score**	C_{kappa}	**MCC**
S1	87.97 ± 18.13	92.11 ± 12.61	90.35 ± 14.89	91.11 ± 13.67	79.87 ± 17.23	81.75 ± 12.70
S2	98.25 ± 2.62	98.71 ± 2.77	98.83 ± 1.33	98.75 ± 1.87	95.84 ± 6.26	95.91 ± 6.13
S3	97.62 ± 2.77	98.97 ± 1.03	97.87 ± 2.79	98.40 ± 1.79	93.47 ± 8.38	93.61 ± 8.11
S4	96.09 ± 3.27	98.39 ± 3.13	95.79 ± 2.22	97.06 ± 2.51	91.21 ± 7.23	91.33 ± 7.24
S5	96.10 ± 4.05	98.82 ± 0.89	95.78 ± 5.15	97.21 ± 2.94	90.72 ± 9.23	91.12 ± 8.35
S6	94.53 ± 1.67	97.71 ± 3.34	94.39 ± 3.0	95.93 ± 1.29	87.57 ± 3.65	88.05 ± 3.35
S7	97.41 ± 1.23	98.74 ± 1.89	97.58 ± 1.28	98.14 ± 0.89	93.82 ± 2.96	93.94 ± 2.90
S8	72.20 ± 11.97	82.96 ± 10.15	76.50 ± 8.89	79.54 ± 9.18	40.58 ± 18.51	41.41 ± 18.34
S9	97.38 ± 1.51	98.70 ± 1.29	97.50 ± 1.84	98.09 ± 1.08	93.94 ± 3.56	94.02 ± 3.50
S10	98.73 ± 1.17	99.60 ± 0.90	98.54 ± 1.19	99.06 ± 0.91	97.04 ± 2.52	97.07 ± 2.51

The sensitivity and $F1$-score, all the evaluation metrics, have been found satisfactory for all the compared classifiers. The human GC consists of 60%–70% of the STP and 40%–30% of the SWP. There exists a slight data imbalance distribution in the dataset. Hence, for all the compared classifiers, C_{kappa} and MCC have been estimated. For all the compared classifiers, the average C_{kappa} has been found to be greater than 86. Moreover, the average % MCC has been found as 91.07 ± 8.92, 93.29 ± 8.36, and 86.82 ± 7.31 (p-value < 0.05) for SVM, LDA, and KNN classifiers, respectively. The LDA classifier demonstrates the highest performance among all classifiers, while the KNN exhibits the lowest performance across all performance measures compared. Therefore, when it comes to classifying phases, the LDA classifier is a more favorable choice compared to the SVM and KNN classifiers.

Fig. 7 depicts the confusion matrix of all three classifiers for GP classification. For SVM and LDA classifiers, the misclassification instances for both the phases (stance and swing) have been found to be at par for SVM and LDA classifiers. However, for the KNN classifier, the misclassification instances for the SWP have been found to almost double as compared to the STP. It reflected the suitability of LDA and SVM over the KNN classifier for phase classification.

Fig. 8 depicts a graphical representation of the predicted GP classes and actual GP classes for one trial of subject 1.

SVM		TRUE	
		Stance	Swing
Predicted	Stance	69.62	1.28
	Swing	1.79	27.31

LDA		TRUE	
		Stance	Swing
Predicted	Stance	69.66	1.23
	Swing	1.05	28.05

KNN		TRUE	
		Stance	Swing
Predicted	Stance	67.82	3.08
	Swing	1.81	27.29

Fig. 7. Confusion matrix of all three classifiers for phase classification.

The dotted lines illustrate the predicted GP classes by the respective classifier, while solid lines represent the actual GP classes estimated by FSR sensors. It has been observed from Fig. 8 that in phase classification, two types of error can occur, outlier type (red patch) and continuous type (yellow patch). The first type of error, the outlier type, can be well-eliminated during the postprocessing and will have no adverse effect on the final performance of the system. However, the second type of error, the continuous type, cannot be handled by postprocessing and remains in the system, which further reduces the performance of the system while estimating the gait temporal parameters.

Table V presents the average temporal gait parameters estimated from FSR sensors (actual value) and predicted values from the outcome of different classifiers. SVM and LDA classifiers estimated the temporal gait parameters more accurately as compared to KNN classifiers. Moreover, the variation in estimated parameters has been found to be minimum for the LDA classifier and maximum for the KNN classifier.

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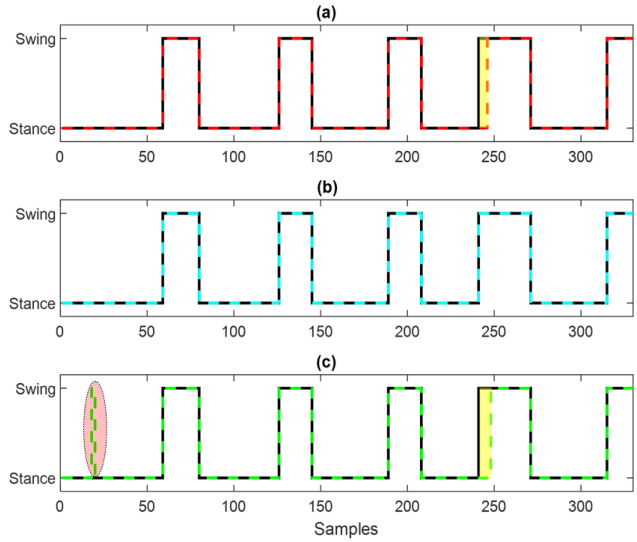

Fig. 8. Graphical representation of the predicted GP classes and actual GP classes for one trial of subject 1. (Red patch: outlier-type error and yellow patch: continuous error). (a) SVM. (b) LDA. (c) KNN.

TABLE V
AVERAGE TEMPORAL GAIT PARAMETERS

Gait Parameters	Actual (FSR)	SVM	LDA	KNN
Stride Time (sec)	0.87 ± 0.11	0.88 ± 0.12	0.88 ± 0.10	0.89 ± 0.09
Stance Time (sec)	0.62 ± 0.02	0.62 ± 0.03	0.62 ± 0.02	0.63 ± 0.08
Swing Time (sec)	0.26 ± 0.04	0.26 ±0.04	0.26 ± 0.05	0.27 ± 0.02
Step Time (sec)	0.43 ± 0.05	0.43 ± 0.05	0.44 ± 0.05	0.44 ± 0.04

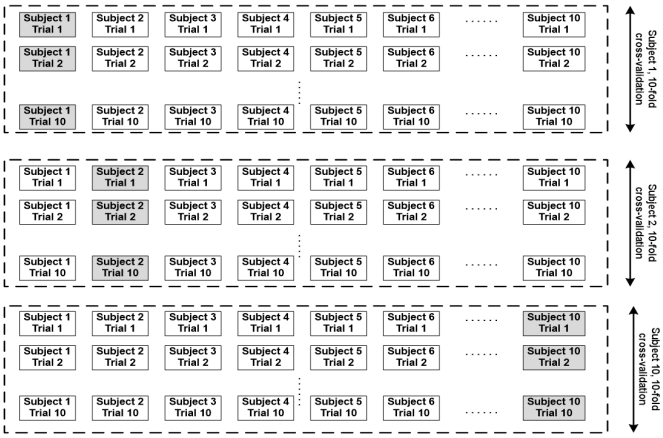

Fig. 9. Methodology for cross-subject performance estimation (gray box depicts the data used to test the classifier).

Further, to gauge the resilience of the proposed model, its performance toward the subject invariability and sensor location invariability has been evaluated. To evaluate the subject variability, the proposed model performance has been evaluated for the naive subject. Fig. 9 depicts the methodology utilized for cross-subject performance estimation. The classifier has been trained for the dataset of all the subjects, excluding the data of the subject to be tested. Fig. 10 shows the % classification accuracy of all the classifiers for GP detection.

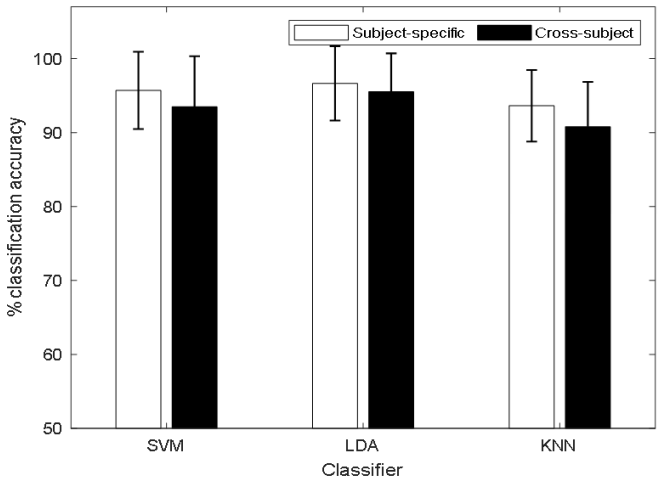

Fig. 10. Subject-specific and cross-subject performance of classifiers for GP classification.

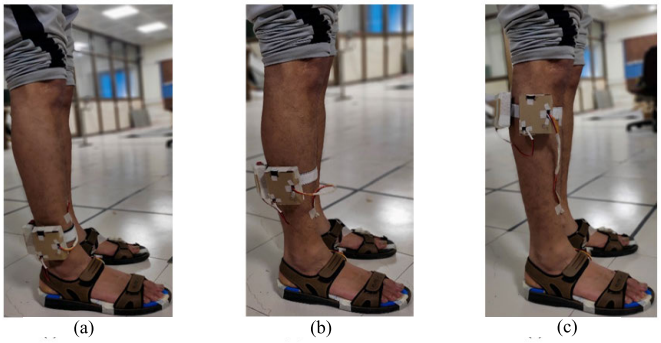

Fig. 11. Different sensor placement locations. (a) Location-1. (b) Location-2. (c) Location-3.

The performance of the KNN classifier has been significantly reduced by 3.03%, and it came down to 90.78% ± 6.05% from 93.62% ± 4.83% (p-value < 0.05). Whereas, for SVM, it has been reduced by 2.30% and estimated as 93.48% ± 6.84% (p-value > 0.05). The best performance is shown by the LDA classifier as 95.52% ± 5.18% with a reduction of only 1.15% (p-value > 0.05).

Further, the proposed model performance for different sensor locations has been estimated for its robustness toward sensor location variability. Fig. 11 depicts different sensor locations used for signal acquisition. Initially, the data acquisition of ten subjects was done by keeping the sensor at location 1 (just above the ankle). However, to further test the sensor location variability on model performance, the data have also been acquired by positioning the sensor at location-2 (in the middle of the shank) and location-3 (just below the knee) separately. The classifiers have been trained with the dataset of location-1 and tested for datasets of both location-2 and location-3 individually. Fig. 12 shows the % classification accuracy of classifiers when the classifier has been trained with a dataset of sensors at location-1 and tested with the dataset of sensors at location-1, location-2, and location-3 separately. The performance of SVM and LDA classifiers has been found consistent for all sensor locations (p-value > 0.05). However,

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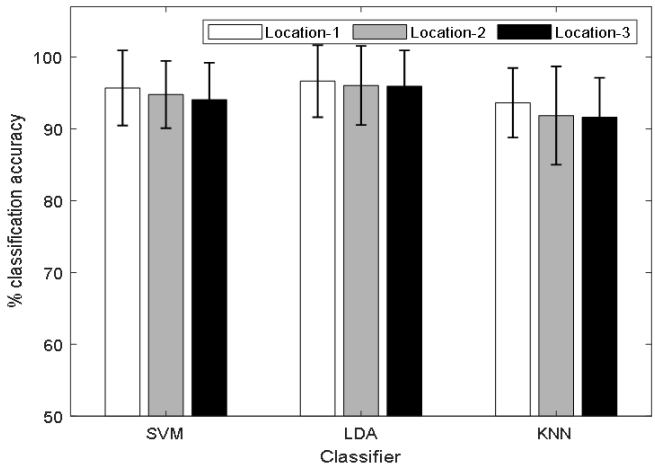

Fig. 12. Classifier performance trained with data of sensor at location-1 and tested with different sensor locations.

TABLE VI
COMPARISON OF THE PROPOSED METHOD WITH OTHER EXISTING METHODS

Authors	Method	Accuracy	Precision	Recall	F1-Score
Liu et al. [54]	HMM	91.88%	91.58%	90.50%	90.95%
Tiwari et al. [3]	CNN	83.45%	-	-	-
Liu et al. [55]	LMM	96.00%	-	-	-
Panahandeh et al. [60]	HMM	95.00%	-	-	-
Dehzangi et al. [40]	CNN	91.00%	-	-	-
Rampp et al. [42]	LSTM	90.00%	-	-	-
Proposed Method	**LDA**	**96.64%**	**97.72%**	**97.45%**	**97.54%**

for the KNN classifier, a slight reduction in performance has been observed (p-value < 0.05)

Table VI presents a comparison of existing techniques and the proposed model in this study. The proposed model performance has been found to be better over all the compared performance indices. Moreover, the proposed model performance has also been found to be invariant to different sensor placement locations, which is not the case with existing techniques.

IV. DISCUSSION

The GP, GE, and its temporal information play a very vital role in rehabilitation, predicting neuromuscular disorder and sleep disorder, and controlling various assistive devices like exoskeleton and lower limb prosthesis. The researchers have proposed various methods. However, the performance of those methods is highly dependent on the controlled environment, calibration of sensors, and sensor positioning. The primary contribution of the proposed method was to develop a portable, cost-effective, user-friendly, robust system to estimate the gait parameters. The proposed system utilized only one IMU sensor and ML to detect GPs and GEs and estimate gait temporal parameters. The developed system consists of one IMU sensor with 6-DoF, a microcontroller, an SD card, and a power supply module. Moreover, a modular insole with four FSR sensors has also been designed and interfaced with the developed module, which was used to validate the performance of the developed module. The developed module utilized the ML approach to predict the GE and GP and estimate temporal gait parameters. Further, a postprocessing strategy was also integrated to filter out the misclassifications. The proposed postprocessing technique makes the GE detection more immune to such type of error, which can also be generated due to involuntary foot movement, foot tapping by poststroke patients [26], subjects suffering from Parkinson's, or any other neuromuscular disorders [52], [53]. Further, the existing, contemporary methods were evaluated under a controlled experimental setup. The performance of the developed module was validated under real-life scenarios with subjects' self-paced walking speeds. Moreover, the developed module utilized the continuous overlapped windowing technique, which makes it suitable for real-time applications.

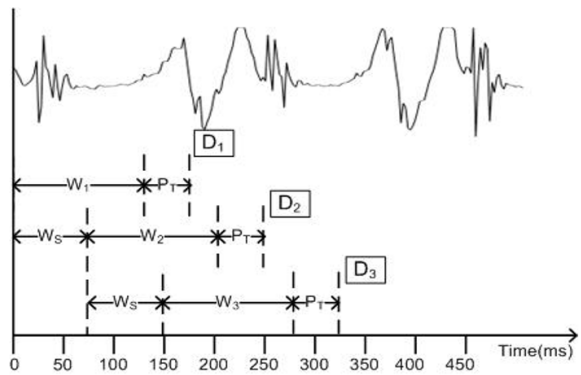

Fig. 13. Continuous overlapping windowing.

Fig. 13 illustrates the overlapping windowing technique with its time complexity. Fig. 13 shows that W1, W2, and W3 are successive windows. For each window, the classification decision (D1, D2, and D3) is made PT seconds later, where PT is the processing time, including the time taken for the feature vector preparation and decision-making by the classifier. The window length specifies the quantity of data utilized for feature extraction and classification, resulting in a single decision. The sliding window WS's step size controls the system's response time between two decisions. Moreover, the processing time PT of the system should be less than or equal to the step size of the sliding window to utilize the developed system's computing capacity fully. In this study, the window length and shift were set to 15 samples (150 ms) and 1 sample (10 ms), respectively. The algorithm's real-time feasibility was evaluated by measuring its computation time using MATLAB R2021b on a Windows 10 system with an Intel[1] Core i9-10900K CPU at 3.70 GHz and 128-GB RAM. The analysis revealed an average processing time of less than 10 ms/GC, demonstrating its potential for seamless real-time deployment. The average processing time for SVM, LDA, and KNN classifiers is found to be 0.0046 ± 0.0232, 0.0023 ± 0.0046, and 0.0044 ± 0.0300 s, respectively. The time delay generated due to the processing time is quite low; hence, it does not

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impact the decisions of the proposed system. Here, to predict the GP, the signal of only a single IMU sensor has been utilized, which eliminates the synchronization issue related to multiple sensors. For each window of the IMU signal, time-domain features have been estimated and concatenated to prepare the feature vector. The proposed methodology is suitable for real-time embedded implementation as it utilizes the continuous overlapping windowing technique, single IMU sensor, and time-domain features.

Table VI provides a comprehensive comparison between the existing techniques [54], [55] and the proposed model introduced in this study. The results demonstrate that the proposed model outperforms the existing approaches across all evaluated performance indices. Furthermore, unlike the existing methods, the proposed model exhibits robustness to variations in sensor placement, maintaining consistent performance regardless of sensor location. In addition, there are other studies that use various other system configurations for gait parameter estimation [3], [56], [57], [58], [59]. Stenum et al. [56] utilized the video-based pose estimation for gait analysis. Stenum et al. [56] proposed a video-based gait analysis method using OpenPose, validated against motion capture in clinical populations. While it enables accessible tracking of gait parameters and rehabilitation progress, its accuracy may be limited by video quality and environmental factors. Dunn et al. [57] demonstrated that video frame interpolation enhances gait parameter precision from single-camera pose estimation, improving accessibility. Lanotte et al. [58] validated the APDM Opal wearable sensor for gait analysis, showing excellent reliability for gait speed, stride length, and timing. However, it underestimated speed and stride length, with errors decreasing at higher speeds and lower asymmetry, warranting caution for slower walking speeds or severe asymmetry. Compared to the above methods, the proposed module offers several advantages over the systems proposed by Stenum et al. [56], Dunn et al. [57], and Lanotte et al. [58] Unlike Stenus et al.'s video-based system, which is sensitive to video quality and environmental factors, the proposed device is more robust and less dependent on external conditions. While Dunn et al.'s frame interpolation improves accuracy, it requires significant computational power, in which the system addresses with efficient machine-learning techniques. Additionally, unlike Lanotte et al.'s wearable sensors, which underestimate gait parameters, the proposed module provides higher accuracy and reliability, even in challenging conditions such as low speeds or high asymmetry.

The developed hardware is not only portable, user-friendly, and easy to wear, but also operates on minimal power. It is position-invariant, calibration-free, suitable for outdoor use, and consumes only 100-mA current when running at a full speed of 600 MHz and operating on 1.25 V. This efficiency makes it a promising solution for long-term use in various applications. The performance for GP prediction has been estimated for SVM, LDA, and KNN classifiers. The average % classification accuracy over ten subjects has been found to be 95.68% $\pm$ 5.22%, 96.64% $\pm$ 5.02%, and 93.62% $\pm$ 4.83% (p-value < 0.05) for SVM, LDA, and KNN classifiers, respectively. The LDA classifier depicts the minimum variation in temporal gait parameter estimation as compared to SVM and KNN classifiers. The LDA classifier outperformed the SVM and KNN classifiers based on all estimated performance measures. Furthermore, the developed module with the LDA classifier depicted promising performance for subject and sensor location variations. In the case of subject variation estimation, the model has been tested for a naive subject.

The system's design, featuring portability, ease of use, and minimal calibration requirements offer substantial advantages for rehabilitation and assistive technologies, setting it apart from many existing systems. Its portability ensures that it can be conveniently transported and utilized in diverse environments, making it significantly more versatile compared to traditional, stationary systems that are often limited to specific settings. The user-friendly interface of this system requires minimal technical expertise, which contrasts sharply with other systems that necessitate extensive training and familiarity with complex interfaces. This ease of use is particularly valuable in both clinical and home-based rehabilitation contexts, where quick setup and immediate functionality are crucial.

A notable technological advancement of this system is its ability to address key issues such as synchronization, sensor placement, and real-time processing with a single IMU sensor. Traditional systems frequently rely on multiple sensors and intricate setups, which can lead to increased complexity and a higher likelihood of errors in synchronization and sensor placement [23], [24]. By streamlining these processes into a single sensor, this system reduces potential issues and simplifies the user experience. This approach not only enhances reliability but also contributes significantly to the system's overall practicality and ease of use, which are essential for its widespread adoption in real-world applications across a large population.

Despite its advancements, the system has limitations that need to be addressed for greater robustness and adaptability. Currently, it has only been validated with a healthy population on uniform terrain, leaving its performance in varied and dynamic environments untested. In contrast, advanced systems often undergo rigorous testing across different terrains and user groups. Moreover, the system lacks subject-specific training, which may affect its performance across users with varying mobility and physical conditions. More sophisticated systems typically use adaptive algorithms to provide personalized support. Additionally, the algorithm's applicability to terrains such as ramps or stairs remains untested, where variations in gait dynamics such as stride length, step timing, and limb coordination could significantly impact its accuracy. Further validation is essential for broader applicability.

Future research should prioritize testing the system in diverse, real-world scenarios and incorporating motion capture systems to validate sensor performance across varying human motion conditions. Long-term studies in these settings will provide a thorough evaluation of the system's capabilities, ensuring that it effectively supports a broader population. Enhancing adaptability and reliability to match advanced systems will be a key focus. Additionally, future work will extend to include uneven terrain, ramps, and stairs, as well as testing on individuals with gait impairments, such as those

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with neurological or musculoskeletal disorders, to evaluate its robustness and clinical applicability.

V. Conclusion

Human gait characterization helps in the treatment and rehabilitation of people with movement/neurodevelopmental disorders. Early assessment of the problem can help in diagnosing/predicting neuromotor disorders at primary stages. However, the current approach for diagnosis is confined to a sophisticated and controlled instrumentation laboratory, which is out of the reach for a large population. The gait characterization module should be user-friendly, cost-effective, robust, and efficient. In this work, an ML-based wearable sensor module has been developed, which consists of only a single IMU sensor. The LDA classifier showed the best classification accuracy for GP prediction as $96.64 \pm 5.02\%$ (p-value < 0.05), in comparison to SVM and KNN classifiers. Further, a post-processing algorithm has been proposed to estimate the GEs from the predicted GPs. The temporal gait parameters have also been estimated from the series of GEs. Stance and swing times measured using the LDA classifier have been found very close to reference values (p-value > 0.05) as 0.62 ± 0.02 and 0.26 ± 0.05 s, respectively. Moreover, the robustness of the proposed model is validated with changing sensor location as well as recruiting naive subjects to predict the GP accurately. The results obtained from this study will help the therapist in the treatment, management, and rehabilitation of gait-associated neurological disorders. In the future, an integrated embedded system powered by deep learning models will be developed for the on-site and off-site implementation of the technology for the telemedicine-based diagnosis, rehabilitation, and management of movement-related disorders.

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References

[1] R. Baker, "Gait analysis methods in rehabilitation," *J. NeuroEng. Rehabil.*, vol. 3, no. 1, pp. 1–10, Dec. 2006.

[2] S. T. Moore, H. G. MacDougall, and W. G. Ondo, "Ambulatory monitoring of freezing of gait in Parkinson's disease," *J. Neurosci. Methods*, vol. 167, no. 2, pp. 340–348, Jan. 2008.

[3] A. Tiwari, A. Pai, and D. Joshi, "A shoe-mounted infrared sensor-based instrumentation for locomotion identification using machine learning methods," *Measurement*, vol. 168, Jan. 2021, Art. no. 108458.

[4] M. Zago et al., "Machine-learning based determination of gait events from foot-mounted inertial units," *Sensors*, vol. 21, no. 3, p. 839, Jan. 2021.

[5] H. F. Maqbool et al., "Heuristic real-time detection of temporal gait events for lower limb amputees," *IEEE Sensors J.*, vol. 19, no. 8, pp. 3138–3148, Apr. 2019.

[6] S. Lahmiri, "Gait nonlinear patterns related to Parkinson's disease and age," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 7, pp. 2545–2551, Jul. 2019.

[7] V. Mikos et al., "A wearable, patient-adaptive freezing of gait detection system for biofeedback cueing in Parkinson's disease," *IEEE Trans. Biomed. Circuits Syst.*, vol. 13, no. 3, pp. 503–515, Jun. 2019.

[8] A. Prado, S. K. Kwei, N. Vanegas-Arroyave, and S. K. Agrawal, "Continuous identification of freezing of gait in Parkinson's patients using artificial neural networks and instrumented shoes," *IEEE Trans. Med. Robot. Bionics*, vol. 3, no. 3, pp. 554–562, Aug. 2021.

[9] Y. Hutabarat, D. Owaki, and M. Hayashibe, "Quantitative gait assessment with feature-rich diversity using two IMU sensors," *IEEE Trans. Med. Robot. Bionics*, vol. 2, no. 4, pp. 639–648, Nov. 2020.

[10] K. Watanabe and M. Hokari, "Kinematical analysis and measurement of sports form," *IEEE Trans. Syst., Man, Cybern. A, Syst., Humans*, vol. 36, no. 3, pp. 549–557, May 2006.

[11] D. Solanki and U. Lahiri, "Design of instrumented shoes for gait characterization: A usability study with healthy and post-stroke hemiplegic individuals," *Frontiers Neurosci.*, vol. 12, p. 459, Jul. 2018.

[12] S. B. O'Sullivan, T. J. Schmitz, and G. Fulk, *Physical Rehabilitation*. Philadelphia, PA, USA: F. A. Davis, 2019.

[13] B. Toro, C. Nester, and P. Farren, "A review of observational gait assessment in clinical practice," *Physiotherapy Theory Pract.*, vol. 19, no. 3, pp. 137–149, Jan. 2003.

[14] A. Cappozzo, U. D. Croce, A. Leardini, and L. Chiari, "Human movement analysis using stereophotogrammetry: Part 1: Theoretical background," *Gait Posture*, vol. 21, no. 2, pp. 186–196, 2005.

[15] M. Montero-Odasso et al., "Dual-task gait speed assessments with an electronic walkway and a stopwatch in older adults. A reliability study," *Experim. Gerontol.*, vol. 142, Dec. 2020, Art. no. 111102.

[16] M. Kirmizi, Y. S. Sengul, and S. Angin, "The effects of gait speed on plantar pressure variables in individuals with normal foot posture and flatfoot," *Acta Bioeng. Biomech.*, vol. 22, no. 3, pp. 1–8, 2020.

[17] W. Tao, T. Liu, R. Zheng, and H. Feng, "Gait analysis using wearable sensors," *Sensors*, vol. 12, no. 2, pp. 2255–2283, Feb. 2012.

[18] X. Liu, B. Zhou, B. Zhang, and T. Liu, "A potential-real-time thigh orientation prediction method based on two shanks-mounted IMUs and its clinical application," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 1, pp. 696–705, Jan. 2022.

[19] A. S. Alharthi, A. J. Casson, and K. B. Ozanyan, "Gait spatiotemporal signal analysis for Parkinson's disease detection and severity rating," *IEEE Sensors J.*, vol. 21, no. 2, pp. 1838–1848, Jan. 2021.

[20] J. Lecoutere, A. Thielens, S. Agneessens, H. Rogier, W. Joseph, and R. Puers, "Wireless fidelity electromagnetic field exposure monitoring with wearable body sensor networks," *IEEE Trans. Biomed. Circuits Syst.*, vol. 10, no. 3, pp. 779–786, Jun. 2016.

[21] O. Beauchet, F. R. Herrmann, R. Grandjean, V. Dubost, and G. Allali, "Concurrent validity of SMTEC® footswitches system for the measurement of temporal gait parameters," *Gait Posture*, vol. 27, no. 1, pp. 156–159, Jan. 2008.

[22] M. Chen, B. Huang, and Y. Xu, "Intelligent shoes for abnormal gait detection," in *Proc. IEEE Int. Conf. Robot. Autom.*, May 2008, pp. 2019–2024.

[23] J. Pawin, T. Khaorapapong, and S. Chawalit, "Neural-based human's abnormal gait detection using force sensitive resistors," in *Proc. 4th Int. Workshop Adv. Comput. Intell.*, Oct. 2011, pp. 224–229.

[24] A. J. A. Majumder, S. I. Ahamed, R. J. Povinelli, C. P. Tamma, and R. O. Smith, "A novel wireless system to monitor gait using smartshoe-worn sensors," in *Proc. IEEE 39th Annu. Comput. Softw. Appl. Conf.*, vol. 2, Jul. 2015, pp. 733–741.

[25] S. Perry and M. Lafortune, "Influences of inversion/eversion of the foot upon impact loading during locomotion," *Clin. Biomech.*, vol. 10, no. 5, pp. 253–257, Jul. 1995.

[26] S. Yang, J.-T. Zhang, A. C. Novak, B. Brouwer, and Q. Li, "Estimation of spatio-temporal parameters for post-stroke hemiparetic gait using inertial sensors," *Gait Posture*, vol. 37, no. 3, pp. 354–358, Mar. 2013.

[27] C.-C. Yang and Y.-L. Hsu, "A review of accelerometry-based wearable motion detectors for physical activity monitoring," *Sensors*, vol. 10, no. 8, pp. 7772–7788, Aug. 2010.

[28] L. Zhou et al., "How we found our IMU: Guidelines to IMU selection and a comparison of seven IMUs for pervasive healthcare applications," *Sensors*, vol. 20, no. 15, p. 4090, Jul. 2020.

[29] I. H. López-Nava and A. Muñoz-Meléndez, "Wearable inertial sensors for human motion analysis: A review," *IEEE Sensors J.*, vol. 16, no. 22, pp. 7821–7834, Nov. 2016.

[30] K. Ren, Z. Chen, Y. Ling, and J. Zhao, "Recognition of freezing of gait in Parkinson's disease based on combined wearable sensors," *BMC Neurol.*, vol. 22, no. 1, pp. 1–13, Dec. 2022.

[31] M. Tomc and Z. Matjačić, "Real-time gait event detection with adaptive frequency oscillators from a single head-mounted IMU," *Sensors*, vol. 23, no. 12, p. 5500, Jun. 2023.

[32] R. Romijnders, E. Warmerdam, C. Hansen, J. Welzel, G. Schmidt, and W. Maetzler, "Validation of IMU-based gait event detection during curved walking and turning in older adults and Parkinson's disease patients," *J. NeuroEng. Rehabil.*, vol. 18, no. 1, p. 28, Dec. 2021.

Authorized licensed use limited to: Malta College of ArtsScience&Tech. Downloaded on November 24,2025 at 11:31:01 UTC from IEEE Xplore. Restrictions apply.

[33] B. Luo, Z. Wang, D. Wang, L. Chen, and X. Ma, "Algorithm for gait parameters estimation based on heel-mounted inertial sensors," *IEEE Sensors J.*, vol. 24, no. 15, pp. 24723–24736, Aug. 2024.

[34] A. M. Sabatini, "Quaternion-based extended Kalman filter for determining orientation by inertial and magnetic sensing," *IEEE Trans. Biomed. Eng.*, vol. 53, no. 7, pp. 1346–1356, Jul. 2006.

[35] M. Haid and J. Breitenbach, "Low cost inertial orientation tracking with Kalman filter," *Appl. Math. Comput.*, vol. 153, no. 2, pp. 567–575, Jun. 2004.

[36] J. H. Lee, H. Ju, and C. G. Park, "Error analysis of PDR system using dual foot-mounted IMU," in *Proc. E3S Web Conf.*, vol. 94, Jan. 2019, p. 2007.

[37] N. A. Salman, A. A. Dehghani Sanij, M. Khan, H. F. Maqbool, N. Iqbal, and T. Hussain, "Computational model for the recognition of lower limb movement using wearable gyroscope sensor," *Int. J. Sensor Netw.*, vol. 30, no. 1, p. 35, 2019.

[38] J. Camargo, W. Flanagan, N. Csomay-Shanklin, B. Kanwar, and A. Young, "A machine learning strategy for locomotion classification and parameter estimation using fusion of wearable sensors," *IEEE Trans. Biomed. Eng.*, vol. 68, no. 5, pp. 1569–1578, May 2021.

[39] C. A. Bailey, A. Mir-Orefice, T. K. Uchida, J. Nantel, and R. B. Graham, "Smartwatch-based prediction of single-stride and stride-to-stride gait outcomes using regression-based machine learning," *Ann. Biomed. Eng.*, vol. 51, no. 11, pp. 2504–2517, Nov. 2023.

[40] O. Dehzangi, M. Taherisadr, and R. ChangalVala, "IMU-based gait recognition using convolutional neural networks and multi-sensor fusion," *Sensors*, vol. 17, no. 12, p. 2735, Nov. 2017.

[41] J. Hannink et al., "Mobile stride length estimation with deep convolutional neural networks," *IEEE J. Biomed. Health Informat.*, vol. 22, no. 2, pp. 354–362, Mar. 2018.

[42] A. Rampp, J. Barth, S. Schülein, K.-G. Gaßmann, J. Klucken, and B. M. Eskofier, "Inertial sensor-based stride parameter calculation from gait sequences in geriatric patients," *IEEE Trans. Biomed. Eng.*, vol. 62, no. 4, pp. 1089–1097, Apr. 2015.

[43] A. Bhongade, R. Gupta, and T. K. Gandhi, "Machine learning-based gait characterization using single IMU sensor," in *Proc. Int. Conf. Comput., Commun., Intell. Syst. (ICCCIS)*, Nov. 2022, pp. 263–266.

[44] F. A. Garcia, J. C. Pérez-Ibarra, M. H. Terra, and A. A. G. Siqueira, "Adaptive algorithm for gait segmentation using a single IMU in the thigh pocket," *IEEE Sensors J.*, vol. 22, no. 13, pp. 13251–13261, Jul. 2022.

[45] X. Jiang, K. H. T. Chu, M. Khoshnam, and C. Menon, "A wearable gait phase detection system based on force myography techniques," *Sensors*, vol. 18, no. 4, p. 1279, Apr. 2018.

[46] F. Bugané et al., "Estimation of spatial–temporal gait parameters in level walking based on a single accelerometer: Validation on normal subjects by standard gait analysis," *Comput. Methods Programs Biomed.*, vol. 108, no. 1, pp. 129–137, Oct. 2012.

[47] K. K. Patterson, W. H. Gage, D. Brooks, S. E. Black, and W. E. McIlroy, "Evaluation of gait symmetry after stroke: A comparison of current methods and recommendations for standardization," *Gait Posture*, vol. 31, no. 2, pp. 241–246, Feb. 2010.

[48] T. Gujarathi and K. Bhole, "Gait analysis using imu sensor," in *Proc. 10th Int. Conf. Comput., Commun. Netw. Technol. (ICCCNT)*, Jul. 2019, pp. 1–5.

[49] L. Pepa, F. Verdini, and L. Spalazzi, "Gait parameter and event estimation using smartphones," *Gait Posture*, vol. 57, pp. 217–223, Sep. 2017.

[50] A. Phinyomark, P. Phukpattaranont, and C. Limsakul, "Feature reduction and selection for EMG signal classification," *Exp. Syst. Appl.*, vol. 39, no. 8, pp. 7420–7431, Jun. 2012.

[51] L.-F. Shi, C.-X. Qiu, D.-J. Xin, and G.-X. Liu, "Gait recognition via random forests based on wearable inertial measurement unit," *J. Ambient Intell. Humanized Comput.*, vol. 11, no. 11, pp. 5329–5340, Nov. 2020.

[52] Y. Watanabe and M. Kimura, "Gait identification and authentication using LSTM based on 3-axis accelerations of smartphone," *Proc. Comput. Sci.*, vol. 176, pp. 3873–3880, Jan. 2020.

[53] P. Pierleoni, A. Belli, O. Bazgir, L. Maurizi, M. Paniccia, and L. Palma, "A smart inertial system for 24h monitoring and classification of tremor and freezing of gait in Parkinson's disease," *IEEE Sensors J.*, vol. 19, no. 23, pp. 11612–11623, Dec. 2019.

[54] L. Liu et al., "Ambulatory human gait phase detection using wearable inertial sensors and hidden Markov model," *Sensors*, vol. 21, no. 4, p. 1347, Feb. 2021.

[55] X. Liu, S. Zhang, B. Yao, Y. Yu, Y. Wang, and J. Fan, "Gait phase detection based on inertial measurement unit and force-sensitive resistors embedded in a shoe," *Rev. Sci. Instrum.*, vol. 92, no. 8, Aug. 2021, Art. no. 084708.

[56] J. Stenum, M. M. Hsu, A. Y. Pantelyat, and R. T. Roemmich, "Clinical gait analysis using video-based pose estimation: Multiple perspectives, clinical populations, and measuring change," *PLOS Digit. Health*, vol. 3, no. 3, Mar. 2024, Art. no. e0000467.

[57] M. Dunn, A. Kennerley, Z. Murrell-Smith, K. Webster, K. Middleton, and J. Wheat, "Application of video frame interpolation to markerless, single-camera gait analysis," *Sports Eng.*, vol. 26, no. 1, p. 22, Dec. 2023.

[58] F. Lanotte, S. Y. Shin, M. K. O'Brien, and A. Jayaraman, "Validity and reliability of a commercial wearable sensor system for measuring spatiotemporal gait parameters in a post-stroke population: The effects of walking speed and asymmetry," *Physiolog. Meas.*, vol. 44, no. 8, Aug. 2023, Art. no. 085005.

[59] C. S. T. Hii et al., "Automated gait analysis based on a marker-free pose estimation model," *Sensors*, vol. 23, no. 14, p. 6489, Jul. 2023.

[60] G. Panahandeh, N. Mohammadiha, A. Leijon, and P. Händel, "Continuous hidden Markov model for pedestrian activity classification and gait analysis," *IEEE Trans. Instrum. Meas.*, vol. 62, no. 5, pp. 1073–1083, May 2013.

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